

GROUP BY

Learning Objectives

- Use GROUP BY to categorize data into meaningful groups
- Apply HAVING to filter groups based on aggregate conditions
- Combine GROUP BY with aggregate functions for group-level summaries

Engage (5-7 minutes)

A school principal wants the average score for each class, not just the overall average. GROUP BY divides records into categories, then aggregate functions compute statistics for each group separately.

Explore (10-12 minutes)

Using a Students table with Class, Gender, and Score columns, write queries to: count students per class, find average score per gender, find classes with average score above 75 using HAVING, and list the top-performing class by average score.

Explain (10-12 minutes)

GROUP BY divides rows into groups based on column values. Aggregate functions then operate on each group. HAVING filters groups after aggregation—unlike WHERE which filters rows before grouping. GROUP BY must appear after WHERE and before HAVING in a query.

Elaborate (8-10 minutes)

A company wants to analyze sales by region: total sales per region, number of orders per region, and regions with total sales exceeding 100,000. Write a query grouping by Region using SUM, COUNT, and HAVING with appropriate conditions.

Evaluate (5-8 minutes)

Write a query to find the number of employees and average salary per department, showing only departments with more than 5 employees and average salary above 50,000. Explain the difference between WHERE and HAVING in this context.